1 Which of the following is true about organelles?

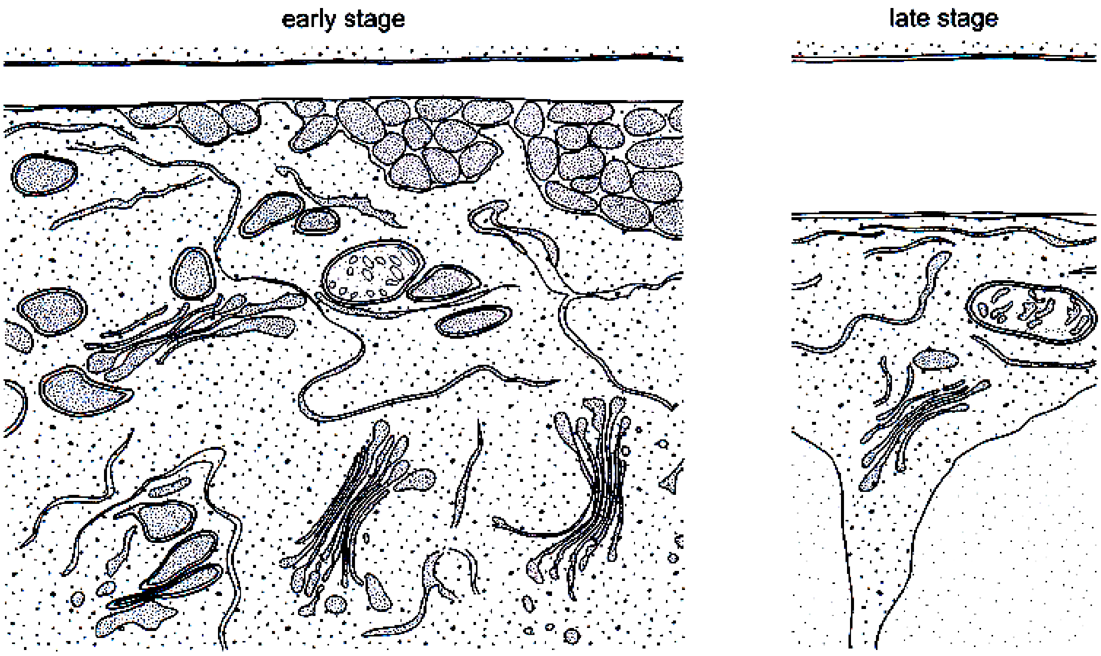
A Vesicles from the smooth endoplasmic reticulum carry lipids to the Golgi apparatus where they are modified.

B Mitochondria and chloroplasts can be seen under a light microscope.

C Mitochondria share the same 80s ribosome as the prokaryotic cell.

D Hydrolytic enzymes are found in lysosomes in animal and plant cells.

2 The photo electron micrographs show early and late stages in the development of the cell wall in a young plant cell.



Which statement describes the events leading to the development of the cell wall?

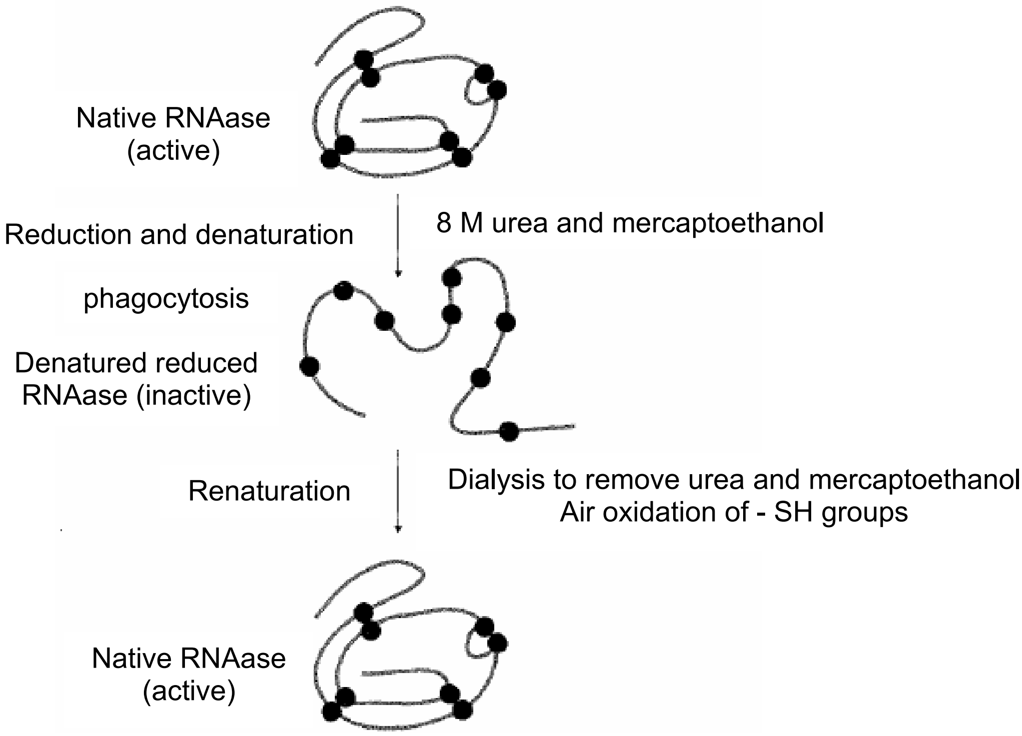
A Complex carbohydrates assembled in the Golgi body are exported to the cell wall by the Golgi vesicles.

B Enzymes in the cell surface membrane synthesise the cell wall components from soluble carbohydrates brought by the Golgi vesicles.

C Polysaccharides are exported to the cell wall and synthesized into wall components by the Golgi body.

D Ribosomes synthesise glycoproteins that are exported by Golgi vesicles to be used in the cell wall.

3 An experiment was carried out to denature a pancreatic ribonuclease, RNAase as shown below. A reducing agent, mercaptoethanol, and urea were added. Denaturation was observed. Upon removal of urea and mercaptoethanol, renaturation occurred.



The following statements describe how polypeptides achieve their specific folding to give a tertiary structure based on the experiment.

I. The polypeptide chain must have refolded as directed by its primary sequence and that once refolded, the original disulphide bonds must have formed again.

II. Given the right conditions, denatured proteins can be refolded.

III. The tertiary structure of a protein can be predicted from its sequence.

IV. Denaturation alters the primary structure of a protein and destroys its biological activity.

Which of the statements are correct?

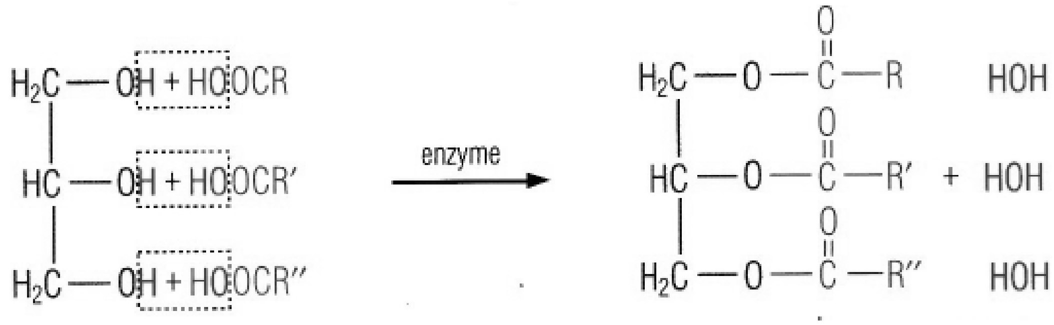
A I and II

B II and IV

C II and III

D I and IV

4 The diagram below shows the formation of a



A Fatty acid

B Glycerol

C Phospholipid

D Triglyceride

5 The flowchart below shows part of an enzyme-catalyzed sequence which involves 4 compounds (**W** – **Z**) and 3 enzymes (***E1*** – ***E3***).

Compound **W** $\xrightarrow{\textbf{E1}}$ Compound **X** $\xrightarrow{\textbf{E2}}$ Compound **Y** $\xrightarrow{\textbf{E3}}$ Compound **Z**

The K_m value of ***E3*** for compound **Y** is 1.1. The K_m value of ***E3*** for compound **Y** in the presence of compound **N** is 1.5. Which of the following statements are correct?

I. Original V_{max} cannot be reached in the presence of compound **N**.

II. Compound **N** is a competitive inhibitor.

III. Binding of compound **Y** alters the globular three-dimensional structure of ***E3***.

IV. The final amount of product formed is the same in both the absence and presence of compound **N**.

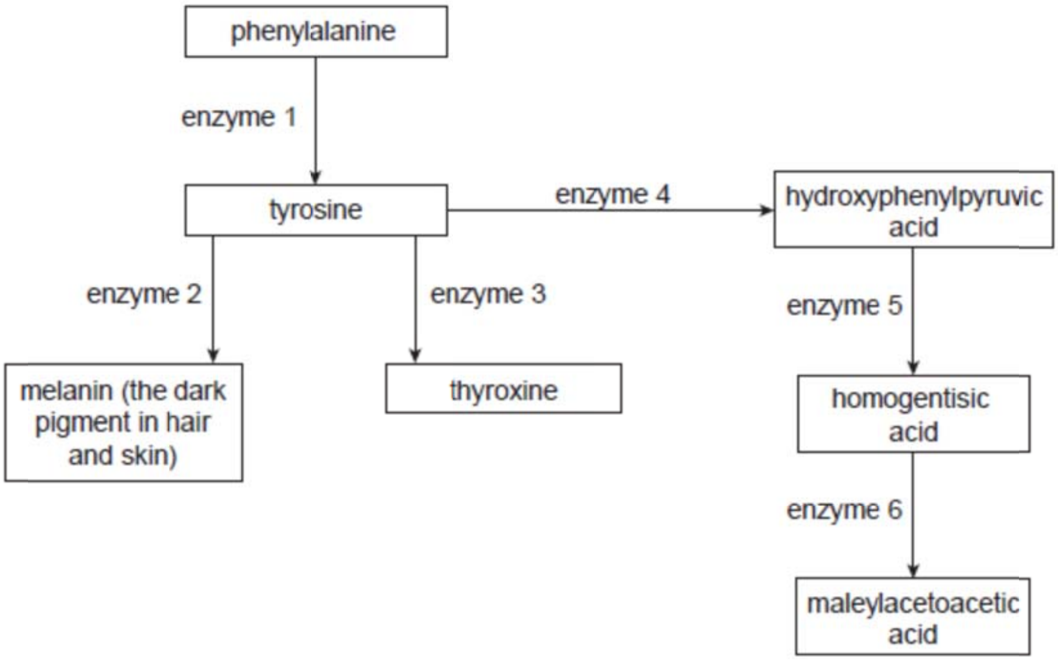
A I and III

B II and IV

C I, II and III

D II, III and IV

6 The following diagram shows part of a metabolic pathway in human cells.



Two genetic metabolic diseases are associated with this pathway: phenylketonuria and alkaptonuria. People with phenylketonuria accumulate phenylalanine. People with alkaptonuria accumulate homogentisic acid. The presence of homogentisic acid in their urine causes it to turn black when exposed to air.

Which one of the following statements is most consistent with the information above?

A People with alkaptonuria lack enzyme **6**.

B People with phenylketonuria lack enzyme **2**.

C People with phenylketonuria and people with alkaptonuria lack thyroxine.

D People with alkaptonuria usually have light-coloured hair.

7 Messenger RNA is isolated by passing homogenised tissue through a fractionating column. The column consists of short lengths of uracil nucleotides attached to a solid support medium. The mRNA that passes through the column cannot be translated as they break up into fragments. The mRNAs that attach to the column can be separated by application of sodium hydroxide.

Which statement is true of the mRNA in this experiment?

I. Complementary base-pairing occurs between adenine and uracil bases in th column.

II. The mRNA that pass through the fractionating column probably lack poly A tails.

III. The mRNA that attach to the fractionating column can be translated.

IV. The mRNA that attach to the fractionating column probably lack poly A tails.

A I and IV

B I, II and III

C II and IV

D All of the above

8 *Escherichia coli*, which has a generation time of 50 minutes, was used in an experiment that showed that DNA replicates in a semi-conservative way.

Which sequence of statements correctly describes the experiment?

1 All bacteria contained ^{15}N DNA.

2 All bacteria contained hybrid DNA (^{15}N DNA and ^{14}N DNA).

3 Bacteria contained either all ^{14}N DNA or hybrid DNA.

4 Bacteria were grown in a ^{15}N medium for many generations.

5 Bacteria were transferred to a ^{14}N medium and sampled every 50 minutes.

A 1 → 2 → 3 → 4 → 5

B 1 → 4 → 5 → 3 → 2

C 4 → 1 → 5 → 2 → 3

D 4 → 3 → 2 → 1 → 5

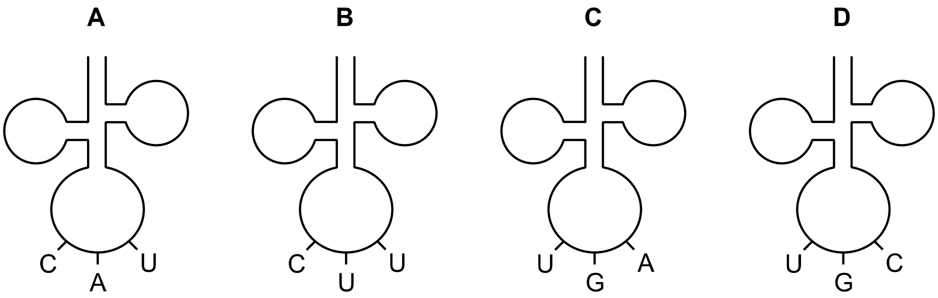
9 Part of the amino acid sequence in normal and sickle cell haemoglobin are shown.

normal haemoglobin	sickle cell haemoglobin
thr-pro-glu-glu	thr-pro-val-glu

Possible mRNA codons for these amino acids are

glutamine (glu) GAA GAG	proline (pro) CCU CCC
threonine (thr) ACU ACC	valine (val) GUA GUG

Which tRNA molecule is not involved in the formation of the part of the amino acid sequence in sickle cell haemoglobin?



10 Genetic drugs are short sequences of nucleotides. One example is an anti-sense drug made from RNA nucleotides that are complementary to the protein-coding mRNA.

Which is the process that is affected by the anti-sense drug and what is its most probable mode of action?

A It prevents DNA replication by binding to the anti-sense strand of DNA that codes for the disease-causing protein

B It prevents transcription by binding to the sense strand of DNA that codes for the disease-causing protein

C It prevents translation by binding to the mRNA that contains information for the disease-causing protein

D It prevents the normal functioning of disease-related protein when it is translated into a protein that inhibits the disease-causing protein

11 The herpes viruses are very important enveloped DNA viruses that cause disease in all vertebrate species and in some invertebrates such as oysters. Some of the human ones are herpes simplex (HSV) I and II, causing facial and genital lesions, and the varicella-zoster (VSV), causing chicken pox and shingles. Each of these three actively infect nervous tissue.

Primary infections are fairly mild, but the virus is not then cleared from the host; rather, viral genomes are maintained in cells in a latent phase. The virus can then reactivate, replicate again, and be infectious to others.

Which process must HSV be able to shut down in order to be able to remain latent in an infected live cell?

A DNA replication

B Transcription of all viral genes

C Apoptosis of a virally infected cell

D All immune responses

12 Which of the following best reflects what we know about how the flu virus moves between species?

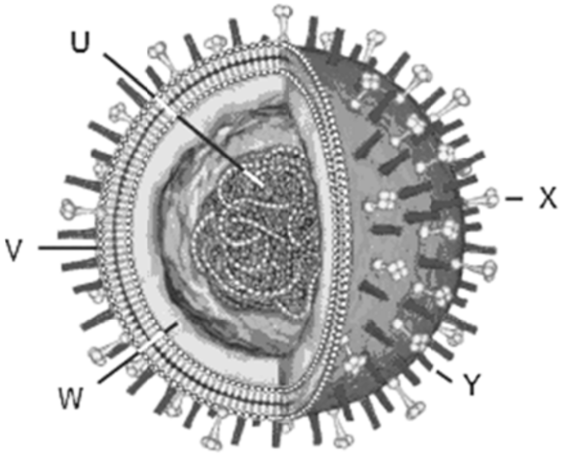
A An avian flu virus undergoes several mutations and rearrangements such that it is able to be transmitted to other birds and then to humans.

B A flu virus from a human epidemic or pandemic infects birds; the birds replicate the virus differently and then pass it back to humans.

C An influenza virus gains new sequences of DNA from another virus, such as a herpesvirus; this enables it to be transmitted to a human host.

D An animal such as a pig is infected with more than one virus and genetic recombination occurs. The new virus mutates and is passed to a new species such as a bird. The virus mutates again and can be transmitted to humans.

13 The figure below represents the structure of the influenza virus.



Which combination correctly identifies the functions of the lettered components?

	U	W	X	Y
A	serves as template for synthesis of complementary RNA strand	protects the viral genetic material	release of newly produced viral particles from the host cell	attaches virus to target cells
B	serves as template for synthesis of complementary DNA strand	protects the viral genetic material	release of newly produced viral particles from the host cell	attaches virus to target cells
C	serves as template for synthesis of complementary RNA strand	fuses with membrane of target cells	protects the viral genetic material	release of newly produced viral particles from the host cell
D	serves as template for synthesis of complementary DNA strand	fuses with membrane of target cells	attaches virus to target cells	release of newly produced viral particles from the host cell

14 For intracellular infectious bacteria and viruses to successfully invade a cell, they must bind to receptors on the cell surface. HIV specifically infects helper T cells, which express the CD4 molecule, but not CD8 on their cell surface, making it possible to distinguish helper T cells from other lymphocytes. Thus, CD4 is hypothesized to be a receptor for HIV.

It is known that HIV cannot infect mice, although the mouse has CD4-positive helper T cells, because mouse CD4 cannot bind to HIV. To study further the mechanism of HIV infection in human cells human genes for CD4 and a human chemokine receptor CXCR4 were expressed in mouse cells.

The results were as follows:

- When the human CD4 gene alone was expressed in mouse T cells, HIV could bind to the cells but could not infect them.
- When human chemokine receptor (CXCR4) was expressed in addition to human CD4 in mouse cells, HIV was able to infect the cells.
- When human CD4 and CXCR4 genes were expressed in mouse cells and the cells were cultivated in the presence of SDF-1a, a ligand of CXCR4, infection by HIV was perturbed.

Which of the following sentences states the best conclusion based on the above experiments?

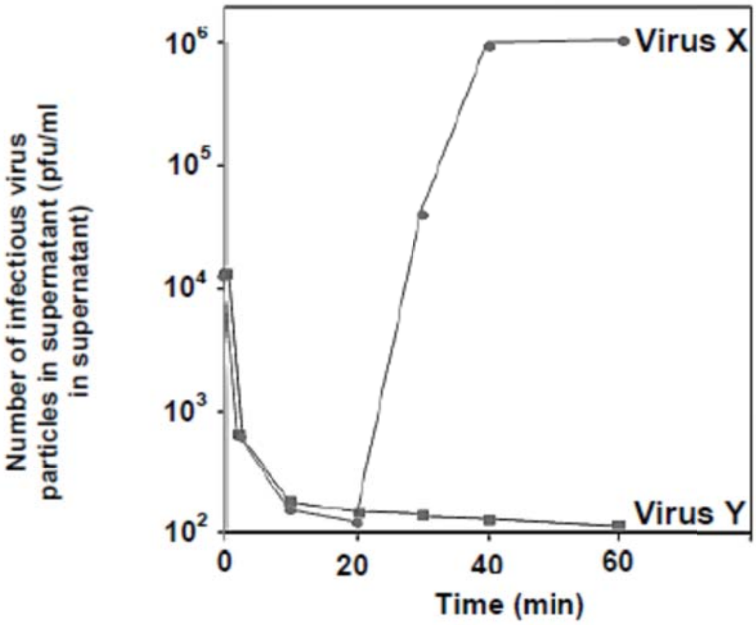
A Even if human CD4 is expressed in mouse T cells, CXCR4 is required for binding of HIV to the T cells.

B Human CD4 is required for the binding with HIV, and the binding is enhanced by the SDF-1a ligand.

C Human CD4 is required for the binding with HIV, but infection of HIV into cells requires help of CXCR4.

D If CXCR4 is expressed in mouse cells, CD4 is not required for the infection of HIV.

15 Two types of viruses, **X** and **Y**, were added to a culture of bacteria. For each type of virus, the change in the number of infectious virus particles present in the supernatant (pfu/ml in supernatant) was monitored for 60 minutes and shown in the graph.



What could explain the sharp increase in the number of infectious virus **X** particles present in the supernatant from 20 to 40 minutes?

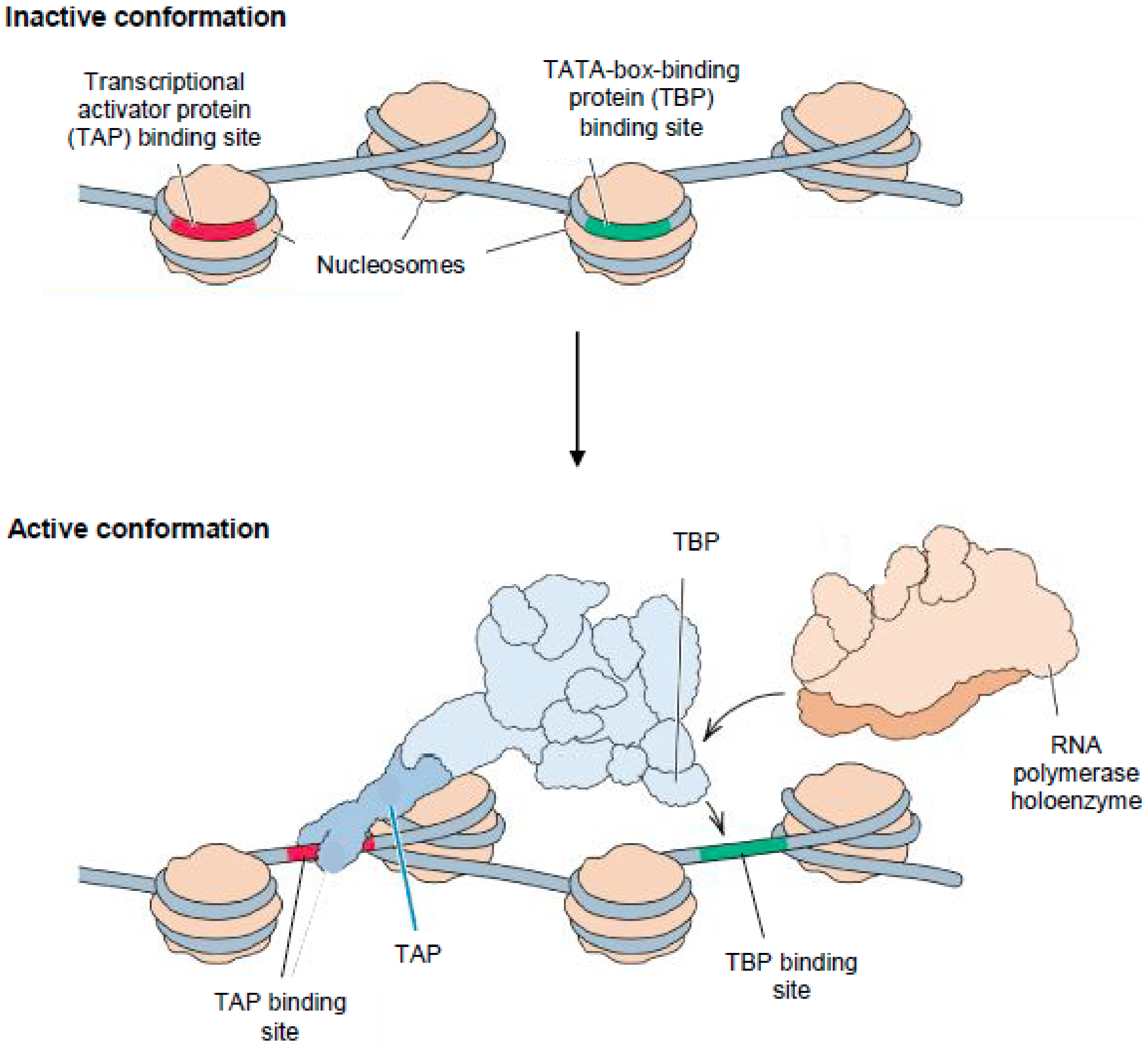
A Injection of viral DNA into host cell

B Integration of viral DNA into host cell DNA

C Release of viral particles by cell lysis

D Release of viral particles by exocytosis

16 The figure shows the process of chromatin modification prior to transcription.



Which of the following statements regarding chromatin modification is correct?

A Chromatin modification complexes result in the reversible acetylation or deacetylation of DNA.

B In the active conformation of chromatin, the repositioning of nucleosomes allows for the exposure of TAP and TBP binding sites, leading to protein binding.

C TAP binding results in the intermediate downregulation of transcription.

D The binding of TAP and TBP to their binding sites is dependent on the recognition of histone tails.

17 The following events occur in the extension of telomeres.

1 Further extension of 3’ end of telomere.

2 Translocation of telomerase to end of telomere.

3 Reverse transcription to extend the 3’ end of the telomere.

4 Complementary base-pairing of RNA template with single-stranded end of telomere.

5 DNA polymerase catalyses formation of complementary strand to form double-stranded DNA.

In which order do these events take place?

A 3 2 4 5 1

B 3 4 2 1 5

C 4 2 3 5 1

D 4 3 2 1 5

18 In the fruit fly, recessive mutations in either of 2 independently assorting genes, brown and purple, prevents the synthesis of red pigments in the eyes. Thus, homozygotes of either of these mutations have brownish-purple eyes. However, heterozygotes for both of these mutations have dark red, that is, wild-type eyes.

Crossing two flies which are both homozygous for a different mutation and heterozygous for the other corresponding mutation produced 144 offspring. How many offspring would be expected to be mutants?

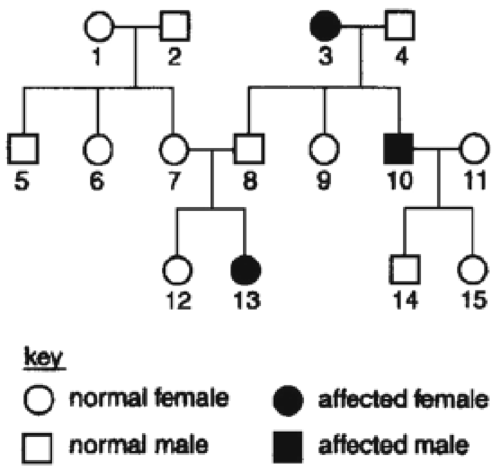
A 63

B 72

C 96

D 108

19 The family tree shows the inheritance of a condition caused by the recessive allele **r**.



Which of the females are certain to have the genotype **Rr**?

A 7, 9 and 15

B 1, 7 and 12

C 1, 6 and 7

D 9, 12 and 15

20 In a small mammal, the allele for grey fur, **G**, is dominant to that for white fur, **g**. The allele for long tail, **T**, is dominant to the allele for short tail, **t**. Animals with grey fur and long tails were crossed with those having white fur and short tails.

The table shows the phenotype of the 55 offspring.

Number of offspring	Fur	Tail
15	Grey	Long
14	Grey	Short
14	White	Long
12	White	Short

Which of the following shows the parental genotypes?

A Ggtt x ggtt

B GGTt x Ggtt

C GgTT x GgTt

D GgTt x ggtt

21 The condition in which a zygote contains three copies of a particular chromosome as a result of nondisjunction is called _______________.

A Trisomy

B Monosomy

C Polyploidy

D Monoploidy

22 In the board bean, a pure-breeding variety with green seeds and black hilums (the point of attached of the seed to the pod) was crossed with a pure-breeding variety with yellow seeds and white hilums. All the F_1 plants had yellow seeds and white hilums. When these were allowed to self-fertilise, the plants of the F_2 generation produced the following seeds.

Yellow seeds and white hilums	89
Yellow seeds and black hilums	28
Green seeds and white hilums	25
Green seeds and black hilums	45

A chi-squared (χ^2) test was performed to test the significance of the difference between the observed and expected results.

$$\chi^2 = \sum\frac{(O-E)^2}{E} \quad \nu = c - 1$$

degrees of freedom	*p = 0.5*	*p = 0.1*	*p = 0.05*	*p = 0.01*	*p = 0.001*
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.36	7.78	9.49	13.28	18.46

Which combination correctly describes the results of the χ^2 test?

	number of degrees of freedom	probability	these are two pairs of segregating alleles at two loci
A	3	> 0.05	Yes
B	3	< 0.05	No
C	4	> 0.05	Yes
D	4	< 0.05	No

23 Fruit flies Drosophila homozygous for long wings, were crossed with flies homozygous for vestigial wings. The F_1 and F_2 generations were raised at three different temperatures.

At each temperature, the F_1 generation all had long wings.

The table below shows the results in the F_2 generation.

Temperature	***Result***
21°C	¾ long wings, ¼ vestigial wings
26°C	¾ long wings, ¼ intermediate wing length
31°C	all long wings

Which statement explains these results?

A Wing length is under polygenic control.

B Long wing and vestigial wing illustrate codominance at 26°C.

C Heterozygous flies have vestigial wings only at 21°C or below but have long wings at 31°C or above.

D Vestigial wing is recessive but causes a vestigial wing phenotype only at lower temperatures.

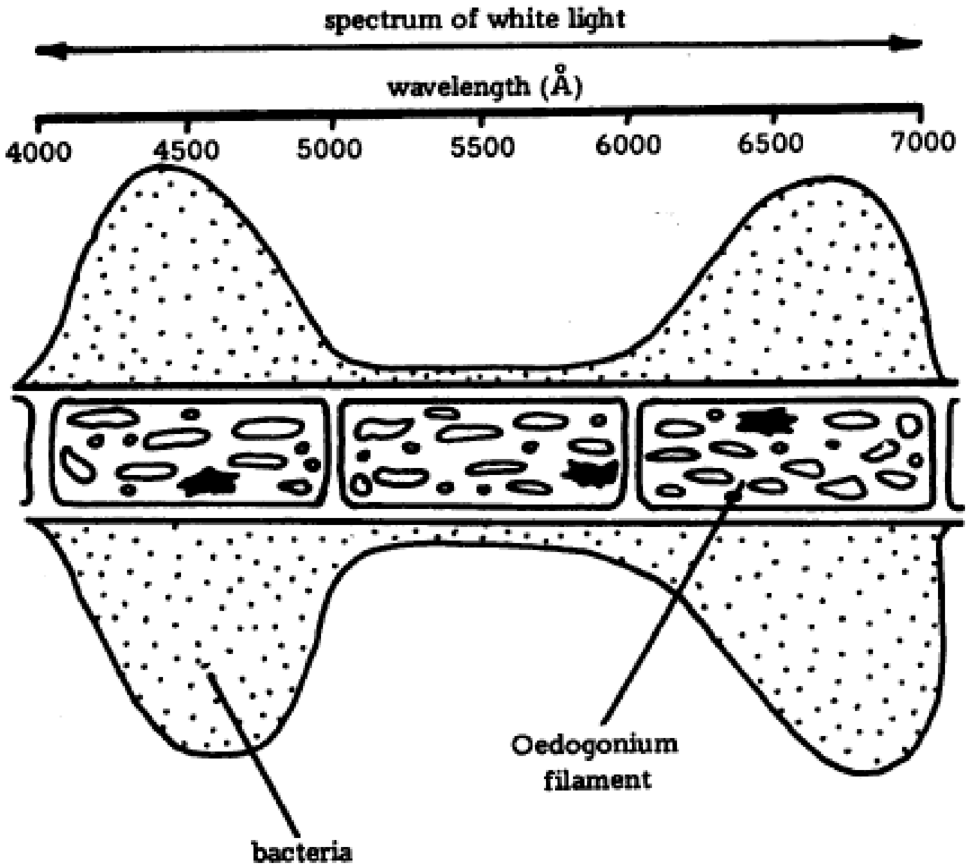
24 The Calvin cycle can be divided into three phases, involving several reactions.

1 carbon dioxide uptake
2 carbon reduction
3 regeneration

Which of the following correctly matched the reactions with three phases?

	ATP hydrolysis	Formation of NADP	Formation of 3C sugar	Use of RuBP
A	1	3	2	1
B	1, 3	2	1, 2	3
C	2	1, 2	2, 3	3
D	2, 3	2	2	1

25 The diagram represents a filament of the green alga Oedogonium sealed in an airtight chamber together with oxygen-sensitive bacteria. The filament was illuminated by a micro-spectrum of white light along the length of three cells. The motile oxygen-sensitive bacteria distributed themselves along the cells as shown.



Which of the following best explains the distribution of the bacteria?

A The bacteria were distributed according to the absorption spectrum of the chlorophyll.

B The central cell had been killed by the blue light and therefore could not attract bacteria.

C The distribution of chlorophyll in the cells was uneven and this influenced the bacteria.

D The two end cells were dead and the bacteria were decomposing them.

26 Which statement about anaerobic respiration is correct?

A Animals are unable to use lactate for the production of ATP.

B From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP.

C From one molecule of glucose, ethanol production yields more energy than lactate production.

D Yeast is able to respire ethanol for the production of ATP.

27 Which of the following correctly describes how a mammal maintains a constant internal environment?

A When there is a deviation from reference point, the detector detects the deviation and transmits the information to the control centre.

B The control centre carries out the response to cancel the deviation, causing a fall or rise back to reference point.

C The effector receives input from the detector and compares the input with the reference point.

D When reference point is reached, there is feedback information and the mechanism is triggered to enhance the deviation in the opposite direction from reference point.

28 A large population of a certain species of freshwater fish lives in South America. Suppose you could somehow cause all mutations to cease in this population and prevent all immigration into the population and emigration from it. Which one of the following statements best expresses the probable future of the population?

A All evolution will promptly cease because without mutation there will be no raw material for evolution.

B The population will begin to deteriorate after 4 to 5 generations because of excessive inbreeding that will result from the absence of immigration and emigration.

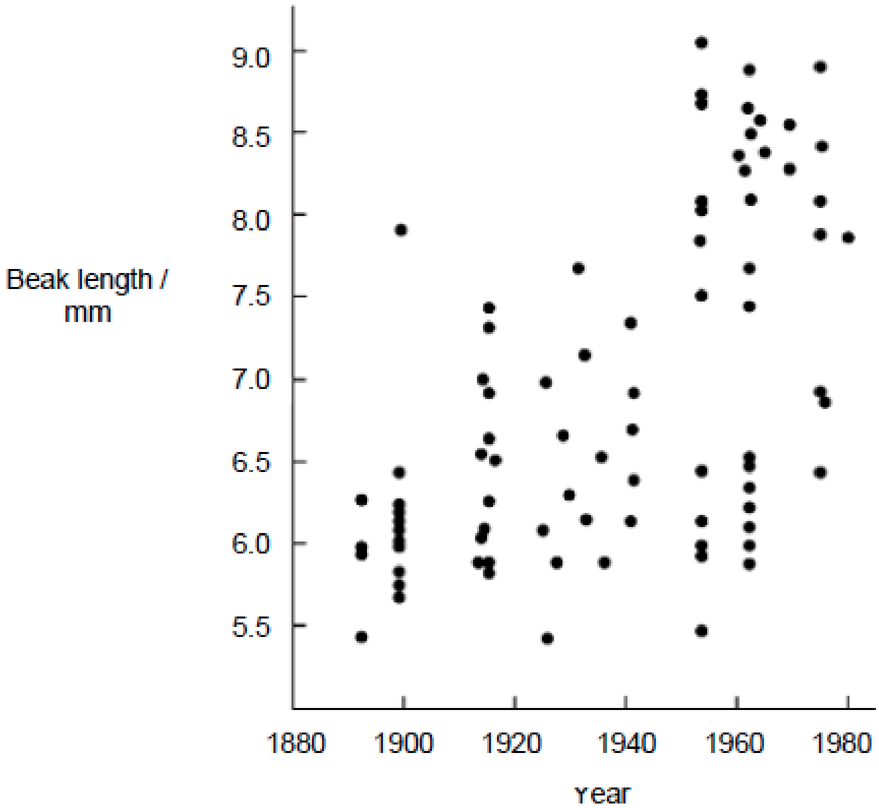
C The population will continue to evolve for a long time as selection acts on the variability produced by recombination.

D Though the population will cease to evolve, it may survive for a long time if the environment remains constant.

Use the following information to answer **Questions 29** and **30.**

The soapberry bug, *Jadera haematoloma*, uses its long beak to penetrate the fleshy fruit of the native soapberry tree to feed on the seeds at the centre. The bug also feeds on the fruit of the introduced golden rain tree.

Investigators measured the beak length of the soapberry bugs over eighty years. The results are shown in the graph.



29 From the information, it would be reasonable to conclude that

A the golden rain tree was introduced around 1970.

B no long-beaked bugs existed prior to the introduction of the golden rain tree.

C the diameter of the golden rain tree fruit acted as a selection pressure on beak length.

D the response of an individual golden rain tree to predation by soapberry bugs would be to grow larger fruit.

30 In the neighbouring regions, the fruits of introduced plants have also been fed on by soapberry buys. Male and female soapberry bugs from different regions can interbreed. Evidence indicates that genetic isolation between some of these populations is gradually occurring.

The situation that would lead to an increase in genetic isolation would be if

A chemicals released by the female bugs attract male bugs from neighbouring populations.

B different types of host plants have fruiting seasons which do not overlap.

C the soapberry tree is common throughout the distribution and each tree produces large amounts of fruit.

D male bugs new to a region are reproductively active, whereas female bugs need to feed before becoming reproductively active.

31 What is one similarity and one difference between classification and phylogeny?

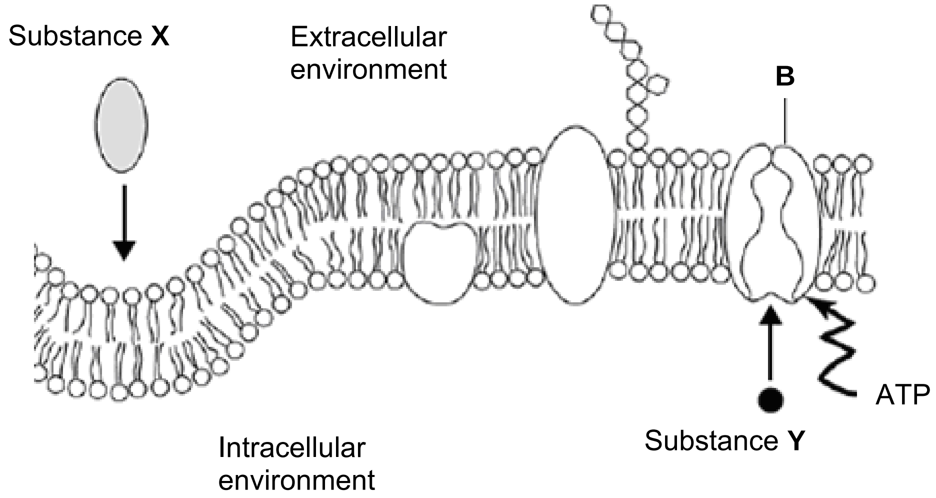
A Both organise organisms into groups based on their characteristics but phylogeny takes into consideration the evolutionary relationships between groups.

B Both put organisms together according to increasingly inclusive groups but classification does not show a branching pattern of evolutionary relationships.

C Both rely on physical characteristics of organisms but phylogeny does not require scientific names.

D Both take into account the common ancestors of organisms but classification is hierachical in nature.

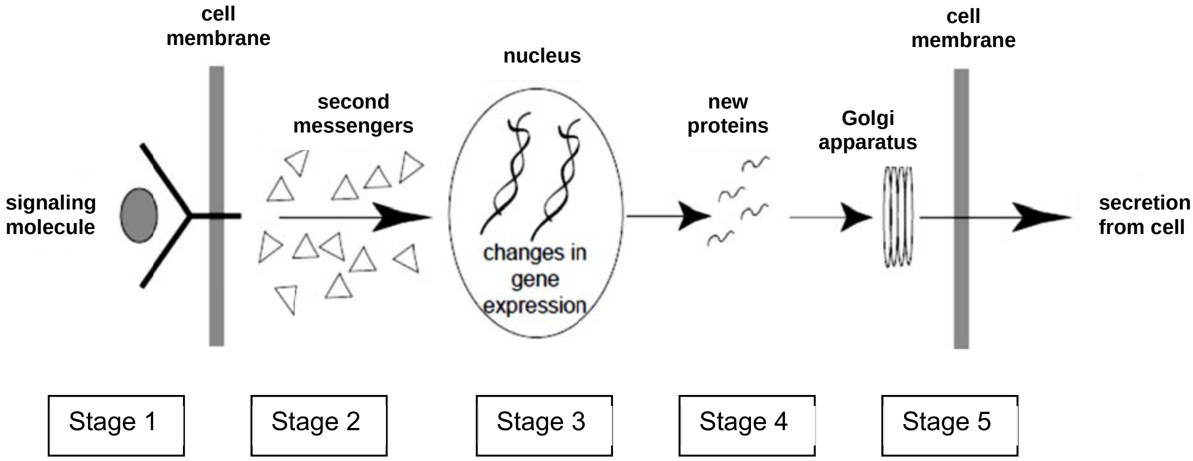
32 The figure below shows a cross section of the plasma membrane in a typical mammalian cell. The substances labelled **X** and **Y** are about to be transported across the membrane in the directions shown by the arrows ($\longrightarrow$).



Which of the following is correct?

	X			Y		
	type of movement	**nature of molecule**	**size of molecule**	**type of movement**	**nature of molecule**	**size of molecule**
A	diffusion	lipid soluble	small	facilitated diffusion	lipid soluble	large
B	diffusion	water soluble	small	active transport	lipid soluble	small
C	endocytosis	lipid soluble	large	facilitated diffusion	charged	large
D	endocytosis	water soluble	large	active transport	charged	small

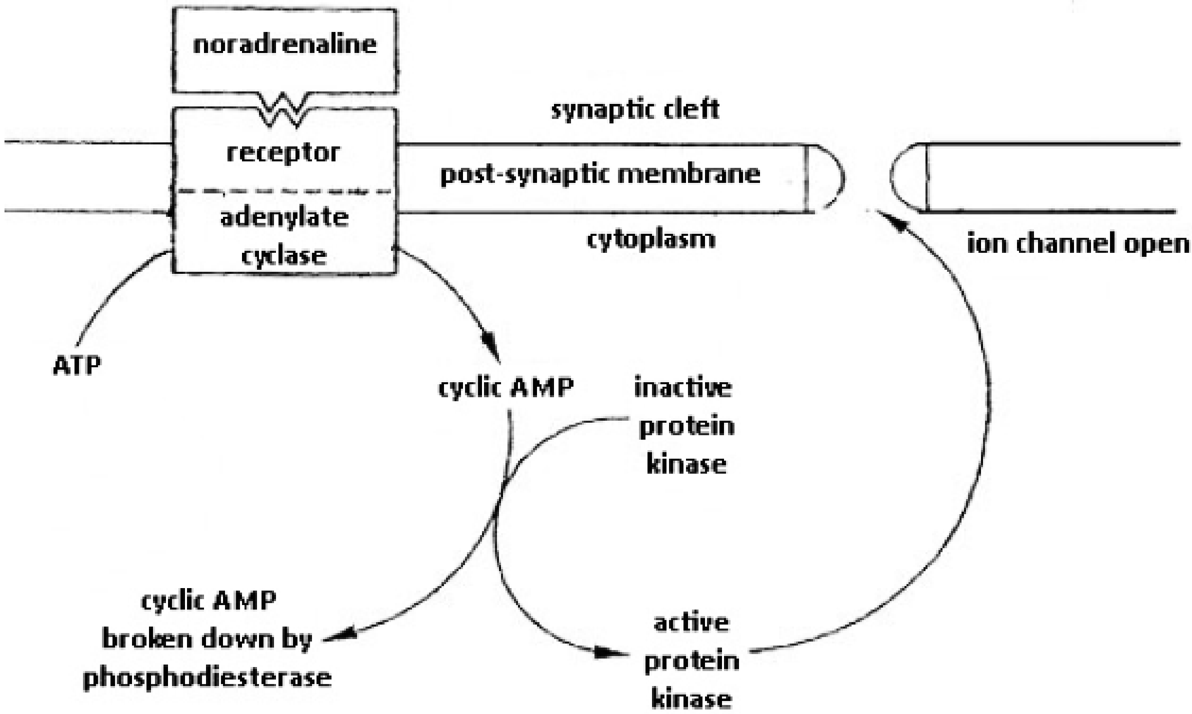
33 The figure outlines the events that occur upon ligand binding.



Which of the following correctly identifies the stages of cell signalling?

	Signal Reception	Signal Transduction	Cellular Response
A	Stage 1	Stage 3	Stage 5
B	Stage 1	Stage 2	Stage 4
C	Stage 2	Stage 3	Stage 5
D	Stage 2	Stage 4	Stage 5

34 Noradrenaline stimulates the activity of an enzyme adenylate cyclase located on the postsynaptic membrane of a neurone. This initiates the sequence of reactions shown in the diagram, causing the opening of channels in the membrane through which ions can pass.



Caffeine causes the ion channels to remain open. A possible explanation for the observed phenomena can be due to the inhibition of

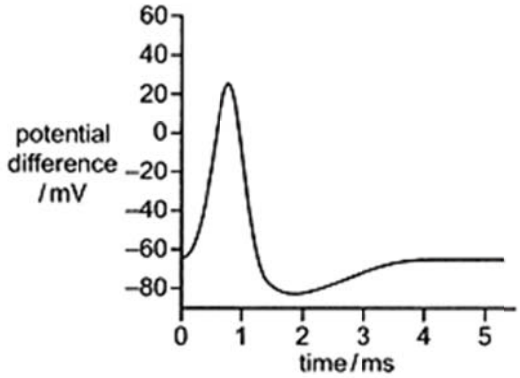
A ATP production

B adenylate cyclase

C cyclic AMP production

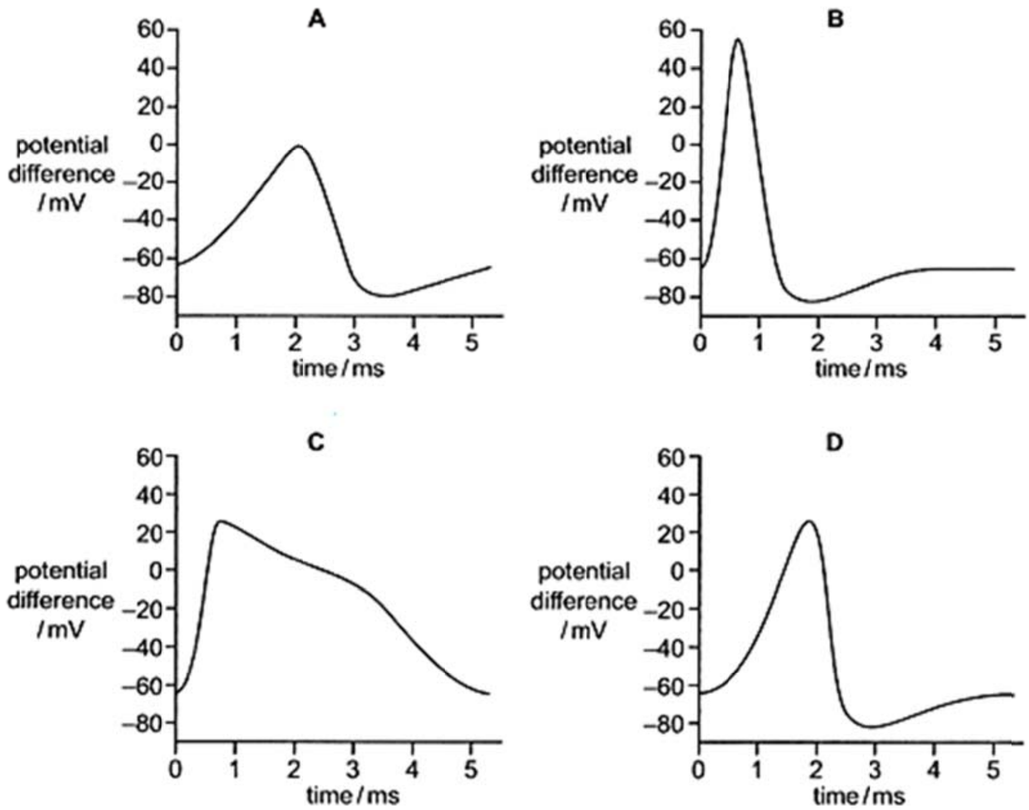
D phosphodiesterase

35 The diagram shows a normal action potential.



Tetraethylammonium ions bind with and block voltage-gated K^+ channels.

Which diagram shows an action potential in a neuron affected by a small quantity of tetraethylammonium ions?



36 What is not a possible issue of concern over the creation of genetically modified farmed animals?

I. Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.

II. Cross species gene transfer may compromise the genome integrity of the species involved.

III. Over production of certain gene products may cause undue stress to the genetically modified farmed animals.

IV. Some genetically modified food products may not be acceptable to certain groups of people.

A I and IV

B II and III

C I, III and IV

D None of the above

37 In mice transfected with the rabbit β-globin gene, the rabbit gene is active in several tissues, including the spleen, brain and kidney. In addition, some of these transfected mice suffer from thalassaemia (a form of anaemia) caused by an imbalance in the coordinate production of α- and β-globins.

Listed below are some problems commonly associated with gene therapy:

I Difficulty in controlling the localisation of the introduced DNA to the target tissue and target location in the genome.
II Difficulty in controlling the output of the introduced DNA.
III Gene expression could be transient.
IV Post-translational modification could be missing.

Which of the above problems are illustrated by the findings on mice transfected with rabbit β-globin gene?

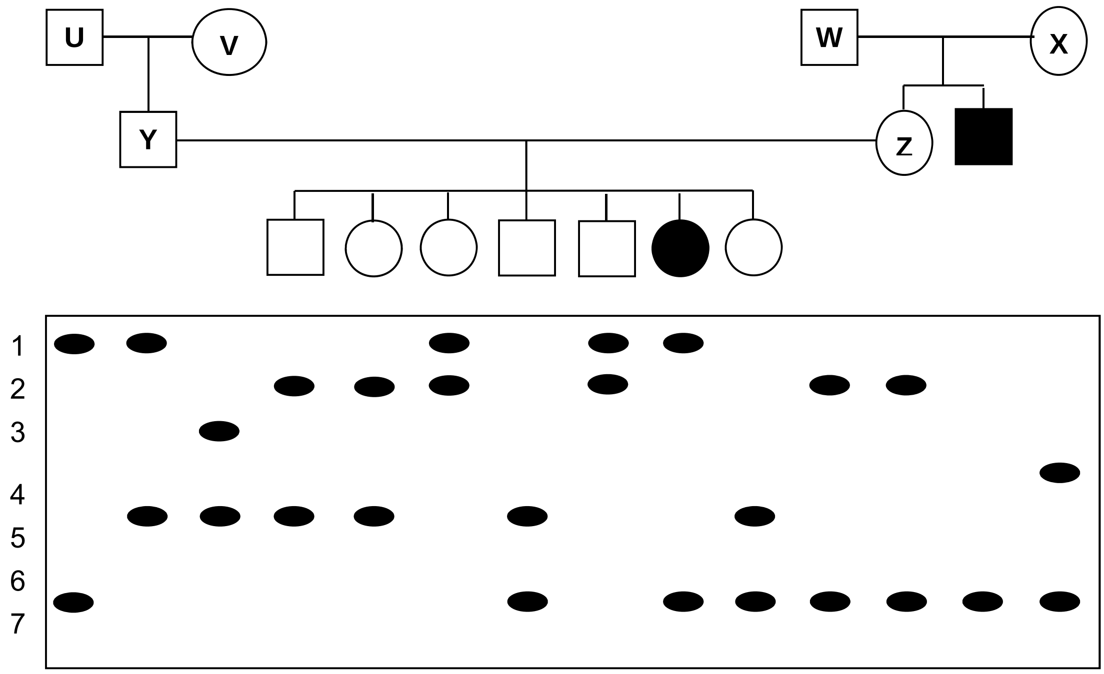
A II and IV

B III and IV

C I, II and III

D I, III and IV

38 A DNA polymorphism with 7 alleles,(1-7), is linked to the gene causing Disease H (an autosomal recessive disease).



In which individual has crossing over taken place between the DNA polymorphism and the gene loci?

A **U** only

B **V** only

C **W** and **X** only

D **Y** and **Z** only

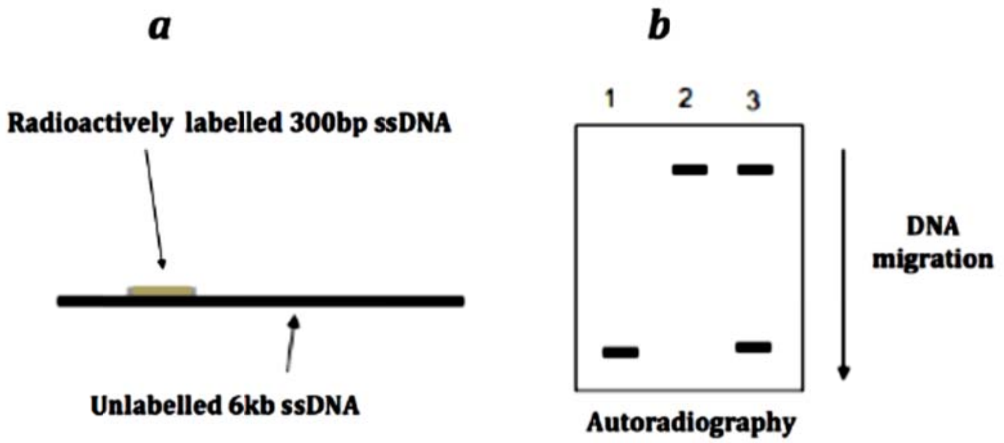
39 DNA helicase, a key enzyme for DNA replication, separates double-stranded DNA into single-stranded DNA. The following describes an experiment to find out the characteristics of this enzyme.

A linear 6 kb ssDNA was annealed with a short (300 bp) complementary ssDNA that is labelled with radioactive nucleotides as shown in ***a***. The annealed DNA was then treated in one of three ways:

- with DNA helicase.
- boiling without helicase.
- boiled helicase.

DNA fragments in the treated samples were separated using gel electrophoresis on an agarose gel. The gel in ***b*** shows the DNA bands that could be detected in the gel by autoradiography.

Note: assume that the ATP required for the reaction has been provided.



Which of the following explanation about this experiment is correct?

A The band appearing in the top part of the gel is the 6.3 kb ssDNA only.

B The band appearing in the lower part of the gel is the labelled 300 bp DNA.

C If the annealed DNA is treated with only DNA helicase and the reaction is complete, the band pattern looks like that in lane 3 in ***b***.

D If the annealed DNA is treated only with DNA helicase and the reaction is complete, the band pattern looks like lane 2 in ***b***.

40 The table shows the functional properties of three thermostable DNA polymerases (Pol) used in the Polymerase Chain Reaction (PCR).

Enzyme	Relative efficiency [a]	Error rate [b]	Processivity [c]	Extension rate [d]	3' to 5' exonuclease activity	5' to 3' exonuclease activity
***Taq* Pol**	88	2×10^{-4}	55	75	No	Yes
***Tli* Pol**	70	4×10^{-5}	7	67	Yes	No
***Pfu* Pol**	60	7×10^{-7}	n.d.	n.d.	Yes	No

[a] Percent conversion of template to product per cycle
[b] frequency of errors per base pairs incorporated
[c] Average number of nucleotides added before dissociation
[d] Average number of nucleotides added per second
n.d. not determined

Which of the following deductions is **not** true?

A *Taq* Pol has proofreading capability.

B *Taq* Pol has the highest relative efficiency as it has a comparatively high processivity and extension rate.

C *Pfu* Pol is likely to have a lower extension rate than *Tli* Pol.

D The lower error rates of *Tli* Pol and *Pfu* Pol can be correlated with their 3' to 5' exonuclease activity.

2014 H2 Biology Preliminary Examination Paper 1 (Answer key)

QN	ANS	QN	ANS	QN	ANS	QN	ANS
1	A	11	C	21	A	31	A
2	B	12	D	22	B	32	D
3	A	13	A	23	D	33	B
4	D	14	C	24	D	34	D
5	B	15	C	25	A	35	C
6	A	16	D	26	A	36	D
7	B	17	D	27	D	37	C
8	C	18	C	28	C	38	B
9	B	19	A	29	C	39	B
10	C	20	D	30	B	40	A